

Marllon Rodrigues

Software & Automation Engineer

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Profile

Engineer with a rare dual track: **industrial SCADA commissioning** on offshore platforms and **full-stack product engineering** for international SaaS. Deployed on-site at Petrobras P80/P83 in Singapore — 50,000+ tags, 1,700+ field devices across two FPSOs. Shipped a cryptographic seal verification system and closed 10+ cross-tenant security vulnerabilities for a Singapore-based maritime platform. Comfortable owning a feature from architecture to production. **English C1.**

Technical Skills

Industrial Automation & OT: AVEVA Plant SCADA (Citect), System Platform, InTouch, Historian, Power Operation (EPO), MES; SEL Relays (F650, 869, 845); Danfoss VFDs; Schneider M580 PLCs; Siemens S7-1200/1500; Rockwell (Control/Compact/MicroLogix); IEC 61850 MMS; Modbus TCP; Profinet; Ethernet/IP; OPC; FPSO Systems; IT/OT Integration; ISA-101.

Software & DevOps: TypeScript, Python, Node.js, PostgreSQL, Prisma, Docker, GitHub Actions CI/CD, C# (.NET), Git, REST APIs, Firestore.

Experience

Automation Analyst, Tyconnex (Singapore/On-site) Feb 2026 – Present

- **FPSO Commissioning (Petrobras P80/P83):** On-site commissioning engineer for AVEVA Plant SCADA (Citect) on two identical offshore platforms — 50,000+ tags, 1,700+ field devices, Schneider M580 PLCs, 168 panels across 5 industrial subnets (MV 13.8kV, 6.6kV, LV, AUX, MULTIPURPOSE).
- **VFD Integration & Troubleshooting:** Commissioned and validated Danfoss VFDs (02A/02B/03A/03B) via Modbus TCP; diagnosed W1007/W1022 faults through live register-level analysis. Commissioned BT19 Ruhrpumpen MV 6.6kV block (SCADA_READ validated, write control complete).
- **TGCP & IEC 61850:** Commissioned TG-5147001 A/B/C/D gas compressors via Modbus TCP over RTDS; resolved critical tag scaling fault. Configured SEL protection relays (F650, 869, 845) via IEC 61850 MMS — 7,000+ signals across 105 IEDs.
- **ESA Network Audit:** Physical survey of 77 switch actions validated across Hirschmann switches (TES001–TES027); identified and reconciled 68 discrepancies including rogue P0 FMM devices.
- **TagFinder Tool:** Built an offline field tool (single HTML file, 4.58MB) — 3,432 P80 + 1,785 P83 equipment records, Modbus address tables, IEC 61850 signal catalog, IP mapping for 1,339 devices; used daily by the field team for real-time troubleshooting.
- **RDO Generator:** Automated vendor daily report pipeline (Python + LaTeX + Docker); 27+ professional PDF reports documenting 231 hours of on-site commissioning.

Full Stack Engineer, Bunkerchain (Singapore/Remote) Jun 2025 – Feb 2026

- **Maritime Platform (MVP 3.0):** Full-stack development on Marine Vessel Pass — a Port Identity Layer for international shipping; FSM-driven port call lifecycle with regulatory holds and multi-signatory workflows. Worked directly with the Singapore founding team.
- **VPS Cryptographic Seal Verification:** Designed and built a trust verification system for physical fuel sample bottle seals — ECDSA digital signatures, public key registry, seal lifecycle state machine, 6 REST API endpoints.
- **RBAC Security Hardening:** Closed 10+ cross-tenant authorization bypass vulnerabilities across 12+ user roles; enforced step-up authentication on sensitive operations; implemented hash-chained audit logs with global sequence integrity.
- **Surveyor Identity System:** Built Licensed Surveyor role with profile, device management, and roster UI; 186+ automated tests for comprehensive RBAC coverage.
- **eBDN Platform (Methanol, delivered):** End-to-end delivery of the Terminal Role on the electronic Bunker Delivery Note system — architecture to live client demo approval.

Systems Developer, MAIMA Technology Aug 2024 – May 2025

- **SAP Automation (RPA):** Python robots automating 6 critical daily SAP processes, reducing manual execution time from 8 hours to under 2 minutes.
- **Predictive Maintenance:** C# system for machine downtime prediction based on product quality analysis, anticipating critical failures before occurrence.

- **CI/CD & Architecture:** Distributed microservices (Image Analysis, Data, BFF layers); AI-powered CI/CD pipelines with automated code review and security scanning on every commit.
- Automation Technician, S.Build Automation

Jun 2022 – Aug 2024
- **Field Engineering & Commissioning:** Specialist in AVEVA System Platform ecosystems, InTouch HMI development, and Historian configuration in critical industrial environments — extensive on-site commissioning travel across Brazil.
 - **Industrial Tech Stack:** Siemens PLCs (S7-1200/1500), Rockwell (Control/Compact/MicroLogix), Schneider. Networks: Profinet, Ethernet/IP, Modbus, OPC. Workflows: AVEVA WorkTasks.
 - **Data Automation & Retrofit:** Python scripts for mass SQL updates in AVEVA MES; legacy migration (1995 → 2023) and reverse-engineering of undocumented machines; upgrade of 20+ servers via ThinManager.
- WorldSkills Competitor (Mechatronics), SENAI Roberto Simonsen

Sep 2021 – Feb 2023
- **State Champion (São Paulo Skills):** Selected as Top 1 student of SENAI Roberto Simonsen for the elite team; team became State Champion. Training: 8–10 daily hours of PLC programming (Siemens S7-1500), precision assembly, Festo MPS stations, and Industry 4.0 / IO-Link integration.

Certifications

- CSI AVEVA Manufacturing Execution System (MES) 2020 – Operations & Performance Exams
- CSI AVEVA Application Server 2023 Exam
- .NET Developer Bootcamp (DIO – 93h) Java OOP (Udemy – 54.5h) Node.js Fundamentals (DIO – 30h)

Education

UNINTER, Systems Analysis and Development	2023 – 2025
SENAI Roberto Simonsen, Computer Networks Technician	2021 – 2022
SENAI Roberto Simonsen, Electronics Maintenance Electrician	2020 – 2022

Languages

- English (C1 – Advanced)
- Portuguese (Native)